Reading Financial Conditions in Units of Monetary Policy

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Abstract

How many points of monetary policy is a movement in financial conditions worth? The answer depends on what the policymaker wants to know. An event study tells how large a conventional policy surprise an observed market movement resembles. A spending model tells how large a policy-rate change would have a comparable effect on GDP. I compare the two for the 10-year yield, equity valuation and the dollar, weighting the assets by their responses to monetary policy news in announcement windows and by their effects on GDP in the FRB/US model. On the combination of asset prices that follows a policy surprise the two agree once they are put on a common scale, and that scale depends on how long the movements are assumed to last. They differ when equities and the dollar move in other proportions: in point estimates the model gives the dollar more weight, relative to equities, than the event study does. Realized unemployment and aggregate GDP growth do not separate the readings, and the tests have little power. In U.S. data, changing the measure of policy news alters financial-condition readings about as much as replacing market weights with spending weights. These choices leave the broad tightening of 2022–24 visible under every calibration, but they change the estimated magnitude, and occasionally the direction, of smaller movements.

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1 Introduction

Suppose the Federal Reserve leaves the policy rate unchanged, but the dollar appreciates and stock prices fall. Both movements tend to restrain spending. How much tightening has occurred? One way to answer is to ask how large a monetary policy surprise would ordinarily produce a comparable movement in financial markets. Another is to ask how large a change in the policy rate would have a comparable effect on spending. A policymaker might reasonably expect the two answers to be close.

Whether they are close depends on which asset prices have moved. Around a typical monetary policy announcement, long yields, stock prices and the dollar move together in fairly predictable proportions. The two methods give similar readings for that combination. But suppose the tightening comes mainly through the dollar. In the Federal Reserve's model, an appreciation of the dollar has a larger effect on spending than its usual response to policy news would suggest. An equity-led tightening is the converse case. The distinction matters whenever policymakers use the familiar effects of their own announcements to interpret financial developments between meetings.

Figure 1 gives an example. Spending-based indexes make such comparisons routine: Caballero and Simsek (2026), using the Goldman Sachs weights, note that a 20 percent fall in equity prices and a 17 percent appreciation of the currency each tighten the U.S. index by about one point, because equities receive a small weight but are volatile and the exchange rate a larger weight through expenditure switching. Here, in one case equity valuations fall by 5 log points. In the other the dollar appreciates by 3.9 log points. I chose the two sizes so that the first method, with policy news measured by a factor from interest-rate futures, rates them alike: each is worth about a quarter of a point of the two-year yield (0.256 and 0.255). The second method, which weights asset prices by their effects on GDP in the FRB/US model, rates the dollar movement 2.51 times the equity movement. The figure also shows the first method when policy news is instead the change in the two-year Treasury yield in the same announcement windows. It then rates the dollar movement 2.06 times the equity movement. How policy news is measured matters as much for this comparison as how spending effects are priced.

Policymakers read the stance of policy from financial conditions. In November 2023 the Chair noted that "Financial conditions have tightened significantly in recent months, driven by higher longer-term bond yields, among other factors" (Powell, 2023), and in March 2024 that "there are many different financial conditions indicators, and you can kind of, you know, see different answers to that question" (Powell, 2024). A reading of financial conditions is a reading of policy only to the extent that a point of the index means a point of policy. This paper asks how many points of policy a movement in financial conditions is worth under two calibrations, how far the answer depends on measurement choices, whether the choice would have changed the reading of history, and whether later activity favors either calibration.

I find that the answer depends on the question. On the usual response of long yields, equities and the dollar to a policy surprise, the two calibrations agree up to a scale that depends on how long asset movements are assumed to last: the model prices the response at 0.954 funds-rate points with persistent paths (interval 0.86 to 1.08) and at 0.82 with a half-life of four quarters, against one by

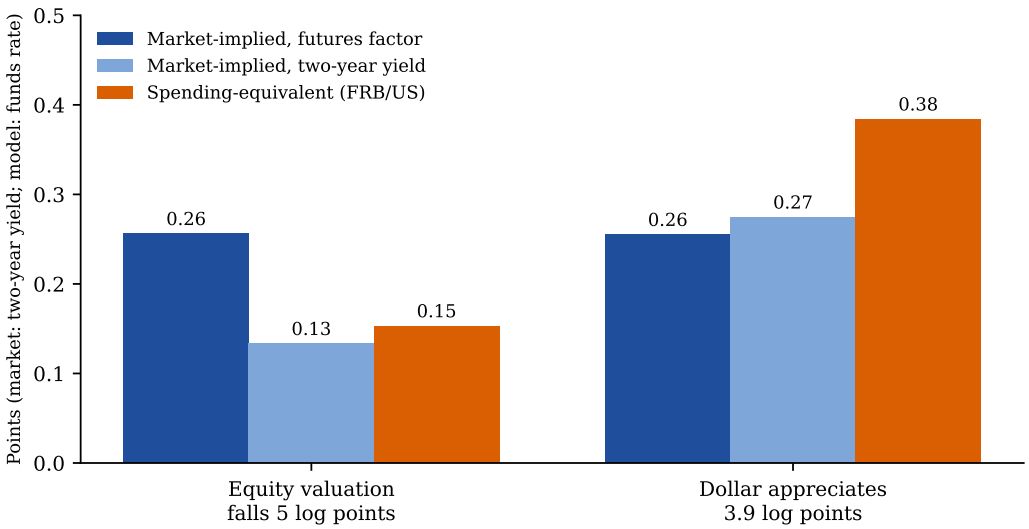


Figure 1: Two Tightenings and Three Ways to Rate Them

Notes: Market-implied readings are in points of the two-year yield, with weights from announcement-window responses to policy news, measured by the futures-based factor (baseline) or by the change in the two-year yield. The spending-equivalent reading is in funds-rate points: the effect on GDP after four quarters of a persistent shift in the variable, divided by the effect of a persistent one percentage point rise in the funds rate. The 10-year yield is unchanged in both cases. Table 3 gives the weights and Table C.1 the intervals.

construction for the event study. They differ in composition. For the two movements of Figure 1 the model rates the dollar movement 0.24 points tighter than the equity movement. The market-implied reading rates them 0.00 apart with the futures factor and 0.14 apart with the two-year yield. In point estimates the model gives the dollar more weight, relative to equities, than the event study does under every definition of policy news that I try, and the size depends on the definition. The measure of policy news matters about as much as the choice of weights: in 12-month changes from 1996 to 2026 the average absolute difference between the model and the baseline market reading is 0.33 of the average market move, and the average difference between the baseline market reading and the market reading with the two-year yield is 0.37.

How often would this have changed an assessment? Across 369 months from 1996 to 2026, the baseline market-implied and spending-equivalent readings of the annual change agree on its direction in 87.5 percent of months. They differ by at least a quarter point in 36.6 percent of months (interval 10 to 70), yet give opposite readings that are both at least that large in only 8 months. The choice changes the size of a tightening much more often than its direction. In 1998–2001 the average readings have opposite signs in the point estimates, both small, and the dollar was the main source of the difference. In 2022–24 every calibration shows a large tightening, about one point or more. Aggregate unemployment and GDP growth do not separate the two readings in the main sample. Composition does predict the net-export contribution to growth there, as the model's channels imply, but not significantly after 1996.

The paper has five parts. First, I define what a policy equivalent is under each method and show why their weights differ (Section 2). Second, I estimate the weights under four definitions of policy news, show how the model's weights arise from expenditure channels, and show how persistence changes the comparison (Section 3). Third, I measure the disagreement on announcements and at horizons from a day to a year (Section 4). Fourth, I examine three historical episodes (Section 5). Finally, I ask whether subsequent activity favors either reading (Section 6). The appendices give the derivations, the construction of the series and the sensitivity of the results.

1.1 Connection to the Literature

Weights and their meaning. Indexes of financial conditions differ mainly in the source of their weights (Hatzius et al., 2010): effects on output from a model (Dudley and Hatzius, 2000; Hatzius and Stehn, 2018; Ajello et al., 2023), principal components (Brave and Butters, 2011), or event-study regressions (Kuttner, 2001; Gürkaynak, Sack and Swanson, 2005; Bernanke and Kuttner, 2005; Nakamura and Steinsson, 2018; Bauer and Swanson, 2023). The older literature on monetary conditions indexes recognized that the weights on asset prices depend on the shock and the dynamics, so that their policy meaning is not invariant (Eika, Ericsson and Nymoen, 1996). I quantify that difficulty with two observed sets of U.S. weights and ask where it matters.

The closest comparisons. Arnaut and Bauer (2024) estimate how an existing index responds to policy surprises, Sack and Swanson (2023) relate FCI-G to policy-induced changes in the two-year yield, and Ajello et al. (2023) translate movements into a growth impulse with model multipliers. Bletzinger, Martorana and Mistak (2026) build a U.S. macro-finance index in units comparable to a policy rate and compare it with the Goldman Sachs index and FCI-G, finding divergences driven by equities and the strongest tightening in 2022–24. I ask how a policy-news weighting and a macro weighting read the same movements. Boehm and Kroner (2025) show that announcements also move equities and exchange rates along a dimension orthogonal to yield changes, so a movement that departs from the conventional pattern can come from the announcement itself.

Information about activity. Hatzius et al. (2010) build an index conditional on past activity and inflation, Aramonte, Rosen and Schindler (2013) show that comparisons among indexes depend on the crisis in the sample, and Lombardi, Manea and Schrimpf (2025) separate safe yields from priced risk and find that risk predicts activity more strongly. Adrian, Boyarchenko and Giannone (2019) show that conditions shift the lower quantiles of growth, and Brave and Butters (2011) build an index of risk, credit and leverage. Caballero, Caravello and Simsek (2026) define FCI*, the level of financial conditions consistent with output at potential. The neutral rate matters for the level of every index (Laubach and Williams, 2003; Holston, Laubach and Williams, 2017, 2023; Hamilton et al., 2016; Bauer and Rudebusch, 2020; Kiley, 2020); I measure the inputs relative to r^* and do not estimate a neutral level for the index (Appendix B). **Why the meaning of a point matters.** Caballero, Caravello and Simsek (2024) study targeting an index of financial conditions in a closed economy, and Caballero and Simsek (2026) extend the analysis to a multi-asset open economy, in which domestic asset prices and the exchange rate both enter aggregate demand and policy

responses link the markets. That gives a reason to care which mix of asset prices delivers a given index reading. My question comes one step earlier: when does a portfolio inferred from conventional policy news give those markets the same relative importance as a spending calibration? The answer does not require endorsing targeting. The same ambiguity arises for anyone who communicates about a desired path of financial conditions, since the path has to say what a point on it means, and for a neutral benchmark such as FCI* (Caballero, Caravello and Simsek, 2026), which takes the aggregation of asset movements as given.

2 Two Ways to Read a Financial Movement

2.1 Two Meanings of a Policy Equivalent

Take a movement in three asset prices: the 10-year Treasury yield, equity valuation and the real broad dollar. Each is measured relative to r^* and to its 1973–2026 average (Appendix B). I call a reading of the movement a policy equivalent if it says how much tightening of the policy rate the movement amounts to. There are two natural ways to compute one.

The market index starts from policy. At each Federal Open Market Committee announcement I measure the policy surprise with the factor of Nakamura and Steinsson (2018), scaled so that one unit raises the two-year yield by one percentage point, and I measure how the three asset prices respond in a narrow window around the announcement. The weights are chosen to recover the size of the surprise from the three asset prices together. A movement is then worth the size of the surprise that would, on average, produce it, in points of the two-year yield. A policy surprise of one unit reads one. I call this the market-implied reading.

The FRB/US index starts from spending. Each asset price is weighted by its effect on GDP four quarters later in the Federal Reserve's FRB/US model (Brayton, Laubach and Reifschneider, 2014), in the experiment in which the asset price shifts persistently, divided by the effect of a persistent one percentage point rise in the federal funds rate. A movement is then worth the funds-rate change with the same effect on GDP, in funds-rate points. I call this the spending-equivalent reading.

The first calculation reads a movement through what policy does to markets. The second reads it through what markets do to spending. Both are legitimate. They need not agree.

2.2 Why the Weights Differ

Consider the dollar. It can be a reliable indicator of a policy surprise because it responds consistently when the expected path of U.S. interest rates changes. How useful it is as an indicator depends on how much it also moves for other reasons, and on what yields and equities already reveal. Its weight in a spending index depends on different things: how an appreciation affects net exports, and how long the appreciation lasts. An asset can be an excellent indicator of a policy surprise and have a modest effect on spending. It can also be a noisy indicator and have a large effect on spending. The same is true of equities, whose effect on spending runs through wealth and the cost of capital. A fall in equity prices that reflects a higher discount rate is policy-like, and a fall that reflects weaker

earnings prospects is not policy at all. The event study averages over announcements, where the first kind dominates. The model prices an equity movement by its estimated effect on spending, whatever its source. Neither calculation is wrong about its own question.

Two equations make the argument precise. Let x be the movement in the three asset prices around an announcement, s the policy surprise, and ψ the vector of responses of the asset prices to a unit of surprise:

$$x = \psi s + u. \tag{1}$$

The vector u collects the ways in which a particular movement departs from the usual response to conventional rate news. It has covariance Σ. Even an announcement can produce such a departure. The market weights w_M recover s from x by generalized least squares, $w_M \propto \Sigma^{-1}\psi$, normalized so that $w_M'\psi = 1$. The FRB/US weights b_h are the effects of the three asset prices on GDP at horizon h in funds-rate points. The model prices the usual response ψ at $b_h'\psi$, which I report in its own units, and dividing by it gives the normalized weights $w_{B,h} = b_h/(b_h'\psi)$, so that both indexes read one on a conventional policy surprise. The difference between the two readings of a movement x is then

$$D_h(x) = (w_{B,h} - w_M)'x = (w_{B,h} - w_M)'u, \tag{2}$$

because the policy part ψs cancels. The same difference has a second reading. The fitted policy component of any movement x is ψM, where $M = w_M'x$ is the market-implied reading; the remainder $x - \psi M$ is the combination of asset prices that a conventional policy surprise cannot reproduce. Once the scales are matched, the spending-equivalent valuation of that remainder is precisely the difference between the two readings,

$$D_h(x) = w_{B,h}'(x - \psi M), \tag{3}$$

because $w_{B,h}'\psi = 1$. A reader can say what the difference measures without assigning it a structural cause. In words, the two methods disagree only to the extent that asset prices move in unusual proportions and the methods value those departures differently. An unusual movement is not enough by itself: if the weights are the same, there is nothing to value differently. A difference in weights is not enough either: if the asset prices keep moving together as they do around policy news, the difference has nothing to act on. That is why two indexes can track each other closely for decades and still give different readings during an equity boom or a dollar surge. Throughout, D is the difference computed on the movements of the three asset prices over the horizon in question, and in Section 6 it is computed on 12-month changes. Appendix A derives the weights.

2.3 Putting Numbers on It

Figure 1 applies the weights to two movements. With policy news measured by the futures factor, the market-implied reading rates a 5 log point fall in equity valuation and a 3.9 log point appreciation of the dollar alike. The spending-equivalent reading gives the dollar movement 2.51 times the effect

of the equity movement, and the ratio of ratios, the model's dollar-to-equity ratio over the market's, is 2.53 (interval 1.08 to 7.41). With policy news measured by the change in the two-year yield, the market-implied reading already gives the dollar 2.06 times the equity movement and the ratio of ratios is 1.22. The market-implied reading is therefore not a single object: its weights depend on what the researcher calls policy news, as the model's depend on the horizon and persistence of the experiment. Neither movement is a policy surprise, so nothing in the construction requires the readings to agree. Whether the difference is large in practice depends on how often asset prices move in such proportions, which is the question of Section 4.

3 Why the Two Answers Agree on a Policy Move

3.1 The Estimated Weights

Table 1 gives the market weights. A policy move that raises the two-year yield by one point raises the 10-year yield by 0.53 points, lowers equity valuation by 7.78 percent and raises the real dollar by 4.22 percent. The data are the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025): 253 announcements from January 1996 to September 2026. Each variable carries a substantial part of the policy move, with shares of 0.33 for the 10-year yield, 0.40 for equity valuation and 0.28 for the dollar. The weights are imprecise, the weight on the 10-year yield most of all (Appendix C.1), and they depend on how policy news is measured (Section 3.2).

The weights say what the variables do when policy moves, not what causes their movements at other times. A rise in stock prices that comes from news about earnings has little to do with policy, and the index treats it as if it did. An announcement can also reveal the Fed's view of the economy, and then the responses mix policy and information (Nakamura and Steinsson, 2018). I measure the responses as markets receive them and set the information effect aside. The index measures policy-equivalent tightening. It is not a forecast of activity.

Table 2 compares the market weights with the FRB/US weights. The two sets agree on the 10-year yield, where the FRB/US weight is 0.92 times the market weight. The model puts less weight on equity valuation (0.60 times) and more on the dollar (1.51 times). On a policy move the

Table 1: Market Index: Responses to Policy News and Weights

	Response	Weight	90% interval	Share of policy move
10-year Treasury yield (pp)	0.53 (0.06)	0.609	[0.147, 1.114]	0.33
Equity valuation, 100 ln CAPE (percent)	−7.78 (0.93)	−0.051	[−0.068, −0.030]	0.40
Real broad dollar, 100 ln (percent)	4.22 (0.53)	0.065	[0.021, 0.112]	0.28

Notes: Announcement windows, 253 announcements, January 1996 to September 2026. Response: change in the window per unit of policy news, which moves the two-year yield by 1 percentage point (heteroskedasticity-robust standard errors in parentheses). Weight: one-factor generalized least squares, normalized so that the weights applied to the responses sum to 1. Interval: 5th and 95th percentiles of weights re-estimated on 5000 bootstrap samples of the announcements. Share: weight times response.

Table 2: FRB/US and Market Weights on the Three Common Variables

	FRB/US weight	Market weight	90% interval	FRB/US response	Market response
10-year yield (pp)	0.563	0.609	[0.147, 1.114]	0.37	0.53
Equity valuation (percent)	−0.031	−0.051	[−0.068, −0.030]	−13.87	−7.78
Real dollar (percent)	0.098	0.065	[0.021, 0.112]	1.56	4.22

Notes: FRB/US weight: effect on GDP after four quarters of a persistent unit shift in the variable, divided by the effect of a persistent 1 percentage point rise in the federal funds rate (0.4616 percent of GDP); funds-rate points per unit. Market weight: two-year-yield points per unit (Table 1); the interval is its bootstrap interval. FRB/US response: change in the variable four quarters after a persistent 1 percentage point funds-rate rise (VAR expectations). Market response: change in the announcement window per unit of policy news.

difference does not matter, because the policy part cancels in equation (2). On a departure from the usual response, the model gives a fall in equity prices its estimated effect on output, and the market index treats it as policy-equivalent.

3.2 What Counts as Policy News?

The market weights are designed to recover a specified signal, so the definition of policy news is a measurement choice in its own right. The baseline signal is the futures-based factor of Nakamura and Steinsson (2018). A natural alternative is the change in the two-year Treasury yield in the same announcement windows, since the index is quoted in two-year-yield points. Table 3 re-estimates the market weights with each definition, using the same 253 announcements for the first two, and also with the factor through 2019 and outside the lower-bound years. The model weights are held fixed and normalized again for each definition.

Two signals on the same announcements, two checks on the sample. The futures factor and the two-year yield are two measures of policy news on the same 253 announcements, and they answer the question of this section. Estimating through 2019 and outside the lower-bound years asks a different question, whether the weights depend on the events used, so I treat these two as checks.

The policymaker's question: the contrast in points. Take the two movements of Figure 1. The model rates the dollar movement 0.24 points tighter than the equity movement (the normalization changes slightly with the signal). The market-implied reading rates them 0.00 apart with the futures factor and 0.14 apart with the two-year yield. The model less market contrast is 0.243 (90 percent interval 0.003 to 0.448) for the baseline, 0.107 (−0.079 to 0.287) for the two-year yield, 0.349 (0.036 to 0.565) through 2019 and 0.232 (0.021 to 0.405) without the lower-bound years. The intervals come from 5,000 bootstrap draws of the announcements; with 500 draws the lower endpoint of the baseline interval was below zero. Switching from the futures factor to the two-year yield lowers the contrast by −0.14 on common draws (−0.27 to 0.03). That is about 14 basis points of the two-year yield for these two movements, and the interval includes zero.

Does the model's dollar-to-equity weight ratio equal the market's? This is a question about proportionality and not about size. The cross-product $c = w_{B,\$} w_{M,E} - w_{B,E} w_{M,\$}$ is zero

Table 3: What Counts as Policy News

	Futures factor	Two-year yield	Factor, through 2019	Factor, no lower bound
Announcements	253	253	201	197
Market weight, 10-year yield	0.609	0.925	0.613	0.986
Market weight, equity valuation	−0.051	−0.027	−0.061	−0.039
Market weight, dollar	0.065	0.070	0.053	0.058
Model reading of the usual response, $b_4'\psi$ [90%]	0.954 [0.856, 1.078]	0.930 [0.834, 1.048]	0.919 [0.805, 1.091]	0.876 [0.790, 0.983]
Market dollar-over-equity ratio [90%]	1.00 [0.33, 2.25]	2.06 [0.82, 5.19]	0.68 [0.09, 2.11]	1.16 [0.38, 2.89]
Ratio of ratios [90%]	2.53 [1.08, 7.41]	1.22 [0.48, 3.05]	3.70 [0.84, 16.30]	2.16 [0.85, 6.16]
Dollar-minus-equity effect, market (points)	0.00	0.14	−0.10	0.03
Dollar-minus-equity effect, model (points)	0.24	0.25	0.25	0.26
Model less market [90%]	0.243 [0.003, 0.448]	0.107 [−0.079, 0.287]	0.349 [0.036, 0.565]	0.232 [0.021, 0.405]
Mean $\|D\|$ over mean |market move|, 12-month changes	0.33	0.36	0.42	0.39
1998–2001, average 12-month change				
Market-implied reading	−0.06	−0.07	−0.14	−0.18
Spending-equivalent reading	0.16	0.16	0.16	0.17
Difference D	0.22	0.23	0.30	0.35
2022–24, average 12-month change				
Market-implied reading	1.02	1.21	1.04	1.30
Spending-equivalent reading	1.01	1.04	1.05	1.11
Difference D	−0.01	−0.17	0.01	−0.19

Notes: The market weights are re-estimated for each definition of policy news: the first principal component of the futures-based factor of Nakamura and Steinsson (2018) (baseline), the change in the two-year yield in the same 253 announcement windows, the factor through 2019, and the factor outside the lower-bound years. The model weights are those of Table 2 in every column, normalized by $b_4'\psi$ for each definition. Effects are for the movements of Figure 1 (equity valuation down 5 log points, dollar up 3.9 log points), market in points of the two-year yield and model in funds-rate points. Intervals are 5th and 95th percentiles over bootstrap samples of the announcements, on common draws. Episode readings are averages of 12-month changes with the weights of each column.

if and only if the two weight vectors give the dollar and equities the same relative weight. Its 90 percent interval excludes zero for the futures factor (−0.0032, −0.0058 to −0.0004) and includes it for the two-year yield (−0.0005, −0.0029 to 0.0015). The ratio of the model's dollar-to-equity ratio to the market's, which tests the same hypothesis on a scale-free ratio, is 2.53 for the baseline, 1.22 for the two-year yield (interval 0.48 to 3.05), 3.70 through 2019 and 2.16 without the lower-bound years. Its denominator, the market's equity effect, is small and imprecise, so I treat the ratio as secondary.

Does switching the signal change the market's own relative weights? The direct test uses the two market weight vectors on the same draws. With F the futures factor and Y the two-year yield, $T = w^F_{M,\$} w^Y_{M,E} - w^Y_{M,\$} w^F_{M,E}$ is zero if the two calibrations give the dollar and equities proportional weights. It is 0.0019 (0.0000 to 0.0037) and positive in 95 percent of the draws. The market's dollar-to-equity weight ratio under the two-year yield is 2.07 times its ratio under the factor (0.98 to 5.47). The change is large in the point estimates, and its interval includes zero at the 10 percent level. Through 2019 and without the lower-bound years the factor's own weights are close to the baseline (0.68 and 1.17 times). I report the change in relative weights as a point-estimate finding. The intervals are not adjusted for the number of definitions.

What each signal captures. The two signals are close in the announcement windows, with a correlation of 0.90, but they are not the same. Write the two-year change as the factor plus a remainder. The remainder is 20 percent of the variance of the two-year change (interval 12 to 30

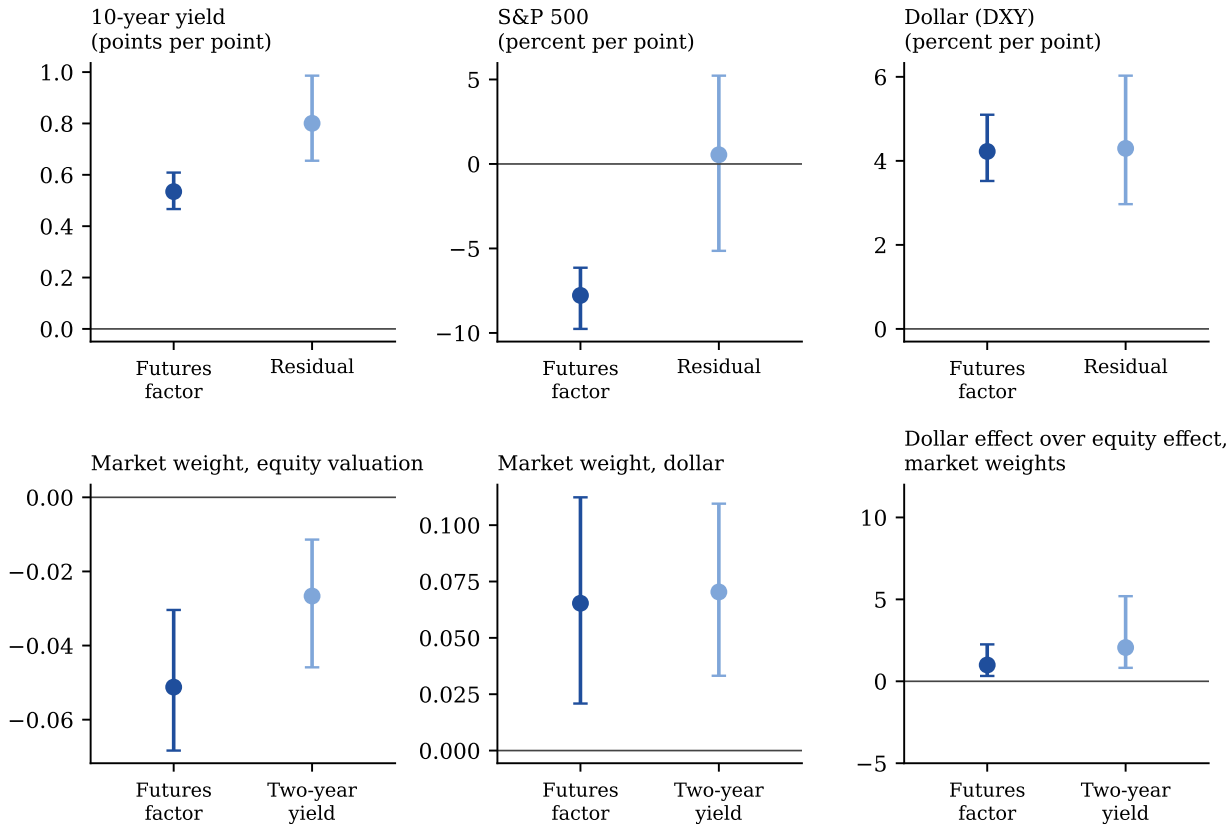


Figure 2: What the Two Measures of Policy News Capture

Notes: Top: responses of the 10-year yield (points), the S&P 500 and the dollar (DXY) (percent) in the announcement windows to one point of the futures factor and to one point of the part of the two-year yield change that is orthogonal to it; the factor is scaled to the two-year change. Bottom: the market weight on equity valuation and on the dollar, and the market's dollar-over-equity effect ratio, when policy news is measured by the futures factor and by the two-year yield. Bars are 90 percent intervals from 5,000 bootstrap draws of the announcements, drawn in common for the two signals.

percent), or about 44 percent of its standard deviation. Figure 2 shows what each part moves. A point of the factor raises the 10-year yield by 0.53 points, lowers the S&P 500 by 7.78 percent and raises the dollar by 4.22 percent. A point of the remainder is associated with a rise in the 10-year yield of 0.80 points and in the dollar of 4.30 percent, and with an equity response of 0.55 percent (−5.14 to 5.23) that the data cannot distinguish from zero. Giving the remainder a larger role in the definition of policy news therefore moves the weights toward the 10-year yield and the dollar and away from equities. The equity weight falls from −0.051 to −0.027. Switching the response vector and the residual covariance in turn, each accounts for about half of the change (0.011 and 0.013), and the order does not matter. The regressions show what variation each signal captures. They do not show whether the remainder is further policy news, a term-premium movement or measurement error.

What carries through. The disagreement is similar under every definition: the ratio of

the mean absolute difference to the mean absolute market move in 12-month changes is 0.33, 0.36, 0.42 and 0.39. The 1998–2001 readings have opposite signs under all four definitions in the point estimates, with the dollar contributing more to the difference than equities (Table 3). In 2022–24 every definition shows a large tightening: both readings are above one point under all four. They are equal to within 0.05 under the baseline and through 2019. Under the two-year yield the market-implied reading is 1.21 and the spending-equivalent reading 1.04, and without the lower-bound years 1.30 and 1.11. That difference comes from the 10-year yield, which the market weights price at 0.93 to 0.99 against about 0.6 in the normalized model weights, and not from equities and the dollar. The near-exact offset of the equity and dollar contributions in 2022–24 belongs to the baseline definition. The judgment that 2022–24 was a large tightening does not.

A check on the spending side. The multipliers of the Board's FCI-G offer a check, with the caveat that they share a lineage with FRB/US, so agreement measures robustness within a family. Per 1 percent change in each variable, the FCI-G dollar multiplier is 2.25 times its equity multiplier, against 3.22 in FRB/US (these ratios use the multipliers per 1 percent, not the sizes of the movements of Figure 1, for which the FRB/US ratio is 2.51). The ratio of ratios against the market is 1.76 under the baseline signal and 0.85 under the two-year yield. FCI-G corroborates the direction under the futures factors and not under the two-year yield. A genuinely independent check would use another model with documented asset-specific multipliers.

3.3 What Lies behind the Model's Weights

The FRB/US weights are effects on GDP, and a reader should see where they come from. Figure 3 follows the two movements of Figure 1 through the model under one persistence assumption, a persistent shift. Only the stated variable moves. The policy rate, the Treasury curve, the mortgage and car-loan rates, the BBB spread, equity valuation, household stock wealth and the dollar are held at baseline in every experiment except their own, so the policy rule is inactive and expectations are those of the model's VAR. The GDP object is the percent deviation of real GDP from baseline. The contributions are the model's own expenditure identities, weighted by shares in GDP, and they sum to GDP.

The equity movement reduces GDP by −0.070 percent after four quarters, and domestic demand accounts for it. Consumption supplies 92 percent of the effect, through wealth and the cost of capital, and business fixed investment 27 percent. Domestic demand in all supplies 120 percent, and net exports offset 33 percent of it, because weaker demand reduces imports; inventories supply 13 percent and government −1 percent, so that the shares sum to 100. The dollar movement reduces GDP by −0.177 percent, and net exports account for 94 percent of it: exports 73 percent and imports 21 percent. Domestic demand contributes 0.6 percent, inventories 13 percent and government −7 percent of the GDP effect. With a half-life of four quarters the shares are similar (consumption 91 percent of the equity effect and net exports 91 percent of the dollar effect). The two weights therefore reflect two different channels, and the dollar's larger weight is a statement about trade. It carries no implication about how persistent the dollar is, which is the assumption of

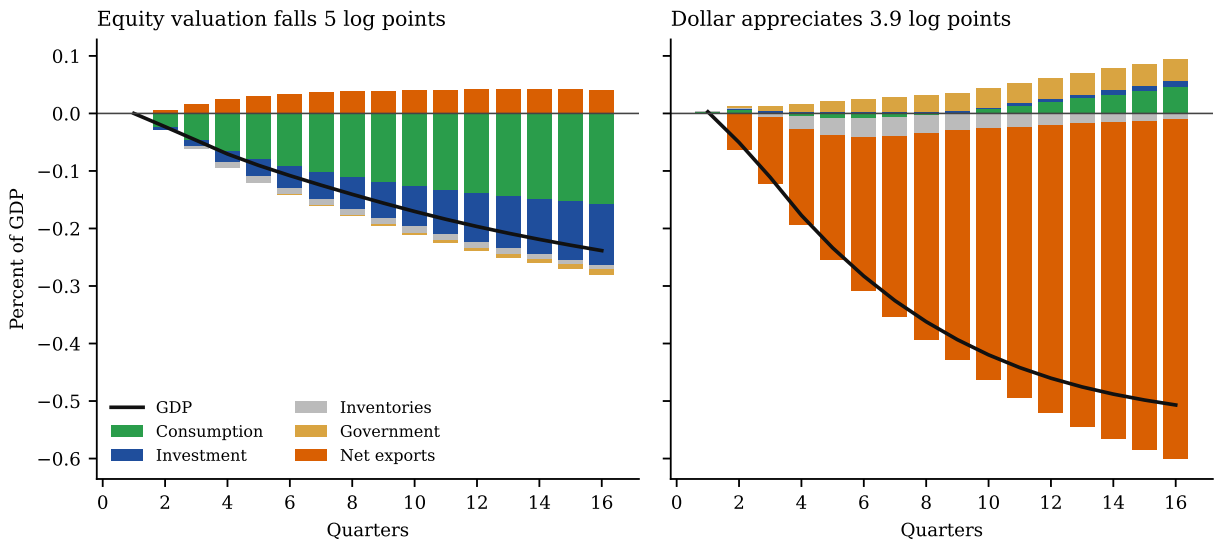


Figure 3: How the Two Movements Reach GDP in FRB/US

Notes: Percent deviation of real GDP from baseline and contributions by expenditure component, persistent shift in the variable, other financial variables held at baseline, policy rule inactive. Investment is residential and business fixed investment; net exports are exports less imports. The figure uses the Board's FRB/US package with the data vintage of October 2026, the vintage behind every FRB/US number in the paper.

the experiment.

The experiments need two mappings from the market series to the model. The equity experiment shocks the required return on equity, and household stock wealth falls with it in the same simulation, so valuation and wealth move jointly; the 5 log point fall in valuation is entered as a 5 percent fall in stock wealth on impact. The market side measures equity by the S&P 500 return in the announcement window, which gives the weight on equity valuation, and applies that weight to monthly changes in the earnings-yield gap converted to CAPE log points (Appendix B); a one percent window return is treated as one log point of valuation, with earnings unchanged. The dollar experiment raises the model's real exchange rate by one percent per unit, and the market side measures the dollar by the DXY in the window and applies the weight to the log of the broad real dollar. DXY is a nominal index of six currencies and the model's dollar is a trade-weighted real index. I treat them as the same unit, and over a window or a month the nominal moves dominate.

The model's own policy shock gives a funds-rate equivalent of 1.77 for the usual announcement mix against 0.95 for the announcement response. FRB/US can therefore assign a value near one to the observed announcement mix even though its own policy shock generates a different mix: heavier in equities and lighter in the dollar. The comparison is not a decomposition of how much of transmission the three assets omit, because the policy shock changes the path, the equity response, the channels that operate and the unit of normalization together. Appendix D.3 gives the experiment.

Table 4: Agreement on Announcements and the Contrast in Composition

FRB/US horizon	Joint response to policy news [90%]	Normalized equity weight	Normalized dollar weight	Dollar over equity, FRB/US	Ratio of ratios [90%]
2 quarters	0.901 [0.81, 1.01]	−0.032	0.089	2.18	2.19 [0.94, 6.44]
4 quarters	0.954 [0.86, 1.08]	−0.032	0.103	2.51	2.53 [1.08, 7.41]
8 quarters	0.917 [0.82, 1.04]	−0.032	0.106	2.57	2.59 [1.11, 7.59]

Notes: FRB/US weights for GDP at the horizon shown, persistent-shift experiment, in funds-rate points; the joint response is the FRB/US weights applied to the responses to a unit of policy news (market index: one). Normalized weights: FRB/US weights divided by that response, so that both indexes read one on a conventional surprise; market weights are 0.609 (10-year yield), −0.051 (equity valuation) and 0.065 (dollar). Ratio of ratios: the FRB/US dollar-over-equity ratio divided by the market's (1.00); the interval comes from the bootstrap of the market weights, and the FRB/US weights are held fixed.

3.4 The FRB/US Reading of a Policy Move

The model prices the usual response of the three variables to a unit of policy news at 0.901, 0.954 and 0.917 funds-rate points at two, four and eight quarters (Table 4). At four quarters the 90 percent interval is 0.86 to 1.08. The market index prices the same response at exactly one. The two do not differ significantly at any horizon with persistent paths; with decaying paths the model's reading is lower (Section 3.5). Nothing in the construction of the FRB/US weights refers to policy announcements, so a value near one is an empirical coincidence in these units. It is not a general property of policy: the model's own policy shock below has a funds-rate equivalent of 1.77 (Section 3.3).

The units differ. A unit of the market signal raises the two-year yield by one percentage point on impact. A unit of the model experiment is a persistent one-point rise in the funds rate. The number 0.954 is a cross-experiment conversion. It does not establish that the two policy paths have the same effect on GDP. Dividing the model's weights by it puts both indexes on one scale and makes them agree exactly along the policy direction. That is an identity and not a finding. The finding is the number before it was divided out.

3.5 Composition or Persistence?

The comparison confounds two things. A persistent dollar appreciation in FRB/US and the dollar's response to policy news are different experiments, so a difference in weights could reflect the initial composition of the movement or the path that follows it. Figure 4 separates them with the same set of calculations under common decaying paths for the three variables, with half-lives of 8, 4 and 2 quarters.

Persistence changes how much tightening a policy move represents. The model's reading of the usual response falls from 0.954 with persistent paths to 0.82 (interval 0.73 to 0.92) with a half-life of four quarters and 0.71 with a half-life of two. Persistence does not change which movement counts more. With the same decay for the dollar and for equities, the model's dollar-to-equity ratio is 2.53, 2.55 and 2.59 at half-lives of 8, 4 and 2 quarters, against 1.00 for the market weights (interval 0.33 to 2.25). Different half-lives for the two assets give a wider range, 1.91 to 3.41, and none near one. Across horizons of two to eight quarters and these paths the model's reading of the usual response

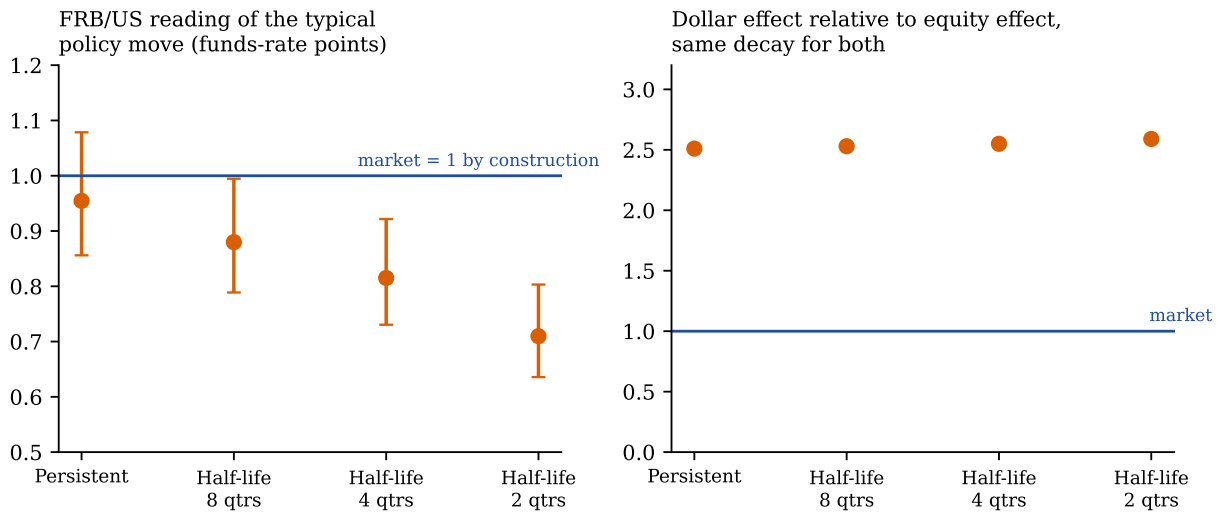


Figure 4: Persistence Changes the Conversion, Not the Relative Weights

Notes: Left: the FRB/US weights at four quarters applied to the responses of the three variables to a unit of policy news, with the variables assumed to persist or to decay with the half-life shown; the bars are 90 percent intervals from bootstrap samples of the announcements. Right: the model's dollar effect over its equity effect, with the same decay for both; the line is the market index's ratio. The persistence of the variables' levels is a different object from the persistence of their responses to policy news; the half-lives of the levels since 1996 are 20 to 26 quarters, and the equity response to policy news does not decay within 20 trading days.

runs from 0.37 to 0.95. The persistent case is the closest to the data, since the half-lives of the levels of the three variables since 1996 are 20 to 26 quarters, so the range for decaying paths shows the sensitivity of the conversion and not a likely value. The contrast between the two weightings is a contrast in the effect on GDP per unit of each asset price. The definition of policy news (Section 3.2) moves the contrast by an amount at least as large.

4 How Much Do the Two Readings Disagree in Practice?

The disagreement depends on how often asset prices move in proportions other than the usual response. I measure it as the mean absolute value of D divided by the mean absolute value of the market move, over the same windows. A ratio of 0.3 means that the typical difference between the two readings is a third of the typical move.

On the 253 announcements, with the weights re-estimated without the event being evaluated, the ratio in intraday windows is 0.19 (interval 0.09 to 0.37), and the 90th percentile of the difference is 0.023 against a mean absolute market move of 0.053. These are the windows in which the weights are estimated, with each event held out.

The ratio is larger in daily data than in the intraday windows and does not shrink at longer horizons. On daily changes of the same three assets, 1996–2026, it is 0.30 on announcement days and 0.32 on other days; at one week it is 0.27 and 0.29, and at 22 trading days 0.27 and 0.25

Table 5: The Difference Between the Readings on Announcement Days and Other Days

Horizon	Announcement windows [90%]	n	Other windows [90%]	n
One day	0.30 [0.17, 0.56]	251	0.32 [0.18, 0.61]	7339
One week	0.27 [0.16, 0.57]	251	0.29 [0.16, 0.59]	1265
22 trading days	0.27 [0.15, 0.56]	248	0.25 [0.14, 0.52]	98

Notes: Mean absolute difference D between the normalized FRB/US reading and the market reading, as a ratio to the mean absolute market move, on the same three assets, 1996–2026 (daily 10-year yield, nominal broad dollar and CAPE; the nominal dollar and no adjustment for r^* stand in for the real dollar and the r^* transformation). Announcement windows end on a scheduled FOMC announcement date; other windows are non-overlapping and contain none. Intervals from joint bootstrap samples of the windows and of the announcements that set the weights.

(Table 5). No difference between announcement and other windows is distinguishable from zero. Announcement days contain a great deal of financial news besides the announcement. The fitted response to policy news is a particular combination of asset prices, and the difference over a whole announcement day looks like the difference on other days.

For windows that end on any day, the ratio is 0.32, 0.29, 0.27, 0.30 and 0.34 at 1, 5, 22, 66 and 252 trading days. A window of 22 or more days contains announcements in any case; removing the moves on announcement days gives 0.33, 0.29, 0.28, 0.32 and 0.35, within 0.02 of those. In 12-month changes of the monthly series, the ratio is 0.33 (interval 0.17 to 0.63), and the difference exceeds 0.25 in 37 percent of months (Figure 5). The months overlap, so runs in this series are not independent evidence of persistence. The statistic is a sample summary with wide intervals. Roughly 0.25 to 0.33 describes the baseline across horizons from a day to a year, and the ratio is 0.36 to 0.42 under the other definitions of policy news (Table 3).

How large is that relative to measurement noise in the market weights themselves? The same statistic computed on bootstrap draws of the baseline market weights against their point estimates has a median of 0.25 (5th to 95th percentile 0.08 to 0.51), and the baseline market weights against those of the two-year definition give 0.37 (0.15 through 2019 and 0.28 without the lower-bound years), against 0.33 for the spending-equivalent weights against the baseline market weights. The disagreement between the model and the market is therefore about as large as the disagreement between two market calibrations that differ only in the measure of policy news (0.37 against 0.33), and it is of the size of the sampling noise in the market weights. The sampling noise limits how finely the two calibrations can be told apart. It is a statement about precision and is separate from the differences across calibrations.

The intraday and daily figures do not isolate a source for the gap between 0.19 and 0.3. The intraday windows are narrow and the weights exclude the event evaluated. The daily comparison uses nominal assets and weights fitted on the same announcements (Appendix D). The comparison across horizons is the cleaner result: the disagreement does not shrink as the window lengthens.

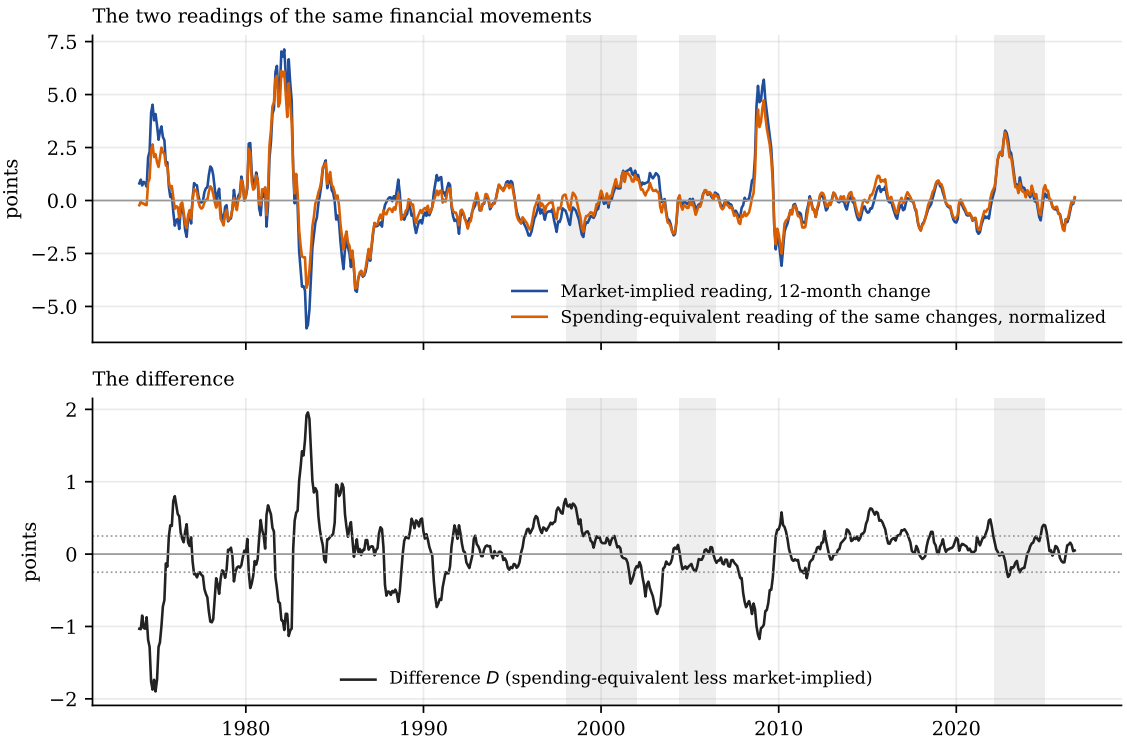


Figure 5: Two Readings of the Same Financial Movements

Notes: Monthly, 12-month changes of the equity valuation, the 10-year yield and the dollar relative to r^*: the 10-year yield less ten-year expected inflation less r^*, the earnings-yield gap to r^* converted to CAPE log points, and the real dollar (Appendix B). Top: the change in the market index and the spending-equivalent reading of the same changes, normalized so that a conventional policy surprise reads one. Bottom: the difference D, FRB/US less market, with dotted lines at ±0.25. Shaded: 1998–2001, 2004–06 and 2022–24.

Table 6: Two Readings of Three Episodes

	1998–2001	2004–06	2022–24
Full indexes, average level (points)			
FRB/US index less market index [90% interval]	1.79 [0.33, 2.97]	−0.11 [−0.76, 0.41]	0.96 [0.00, 1.76]
of which: equity weights	1.32	0.67	0.54
Same three assets, 12-month change (points)			
Market-implied reading	−0.06	0.04	1.02
Spending-equivalent reading	0.16	−0.05	1.01
Difference D	0.22	−0.09	−0.01
Change in the unemployment gap over the next 12 months (points)			
Realized	0.35	−0.41	0.27
Forecast from the market-implied reading	0.67	0.13	0.65
Forecast from the spending-equivalent reading	0.78	0.10	0.66

Notes: Top block: difference in the average levels of the full FRB/US index (with house prices) and the market index relative to r^*, in points, and the part that comes from the equity weights (Table C.2). Middle block: 12-month changes of the market index and of the normalized FRB/US reading, computed on the same three assets and averaged over the episode; D is their difference. Bottom block: change in the unemployment gap over the following 12 months, averaged over the episode's origins, and recursive out-of-sample forecasts of it from the market reading and from the FRB/US reading (own lags included). A gap in levels can coexist with agreement in changes.

5 Would a Policymaker Have Read History Differently?

Three episodes show when the choice of weights matters. All readings are retrospective estimates, made with weights estimated on the full sample. For each episode I report two comparisons that are easily confused. The first is the difference between the full FRB/US index and the market index in average level, which includes variables the market index does not have and the effect of centering each variable on its average. The second applies the two weightings to the same three assets in 12-month changes, which isolates the weights. A gap in levels can coexist with agreement in changes. Table 6 gives both. Figure 6 shows the baseline readings month by month and the contributions of each asset to their difference, and Figure 7 shows the two readings under every combination of the measurement choices of Section 3.

How often would the choice have changed an assessment? Before the episodes, consider the full history. For every month from 1996 to 2026 I read the 12-month change in the three assets with the baseline market weights and with the FRB/US weights on the market's scale. Table 7 counts the months in which the two readings have the same sign, differ in sign only because at least one reading is within a threshold of zero, oppose each other with both readings at least the threshold, and differ by at least the threshold. With a threshold of a quarter point, the readings agree on direction in 87.5 percent of the 369 months. They differ by at least a quarter point in 36.6 percent of months, with an interval of 10 to 70 percent across draws of the market weights, and oppose materially in 8 months (interval 0 to 5 percent). The other 38 months with opposite signs are months in which at least one reading was within a quarter point of zero; the other reading could have been sizable. With a threshold of 10 basis points, 19 months oppose materially. Under the

Table 7: How Often the Two Readings Differ

	1996–2026	1998–2001	2022–24
Months	369	48	34
Threshold t = 0.10 point			
Same sign	323 (87.5%) [78, 94]	41 (85.4%) [75, 94]	33 (97.1%) [76, 100]
Opposite signs, at least one reading within t of zero	27 (7.3%) [5, 10]	3 (6.2%) [4, 12]	1 (2.9%) [0, 12]
Opposite signs, both readings at least t	19 (5.1%) [0, 13]	4 (8.3%) [0, 17]	0 (0.0%) [0, 15]
Readings differ by at least t	269 (72.9%) [48, 88]	41 (85.4%) [54, 96]	23 (67.6%) [35, 91]
Threshold t = 0.25 point			
Same sign	323 (87.5%) [78, 94]	41 (85.4%) [75, 94]	33 (97.1%) [76, 100]
Opposite signs, at least one reading within t of zero	38 (10.3%) [6, 17]	4 (8.3%) [6, 19]	1 (2.9%) [0, 21]
Opposite signs, both readings at least t	8 (2.2%) [0, 5]	3 (6.2%) [0, 6]	0 (0.0%) [0, 3]
Readings differ by at least t	135 (36.6%) [10, 70]	21 (43.8%) [15, 81]	6 (17.6%) [0, 74]

Notes: Monthly 12-month changes in the three market variables (relative to r^*), read with the baseline market weights and with the FRB/US weights on the market's scale (points of the two-year yield). Each cell gives the number of months, the share of months, and the 5th to 95th percentile of the share over announcement-bootstrap draws of the market weights (the 12-month changes are held fixed; months overlap). 1996–2026 is 369 months, 1998–2001 is 48 and 2022–24 is 34.

other definitions of policy news the number of materially opposing months at a quarter point runs from 4 with the two-year yield to 14 through 2019. These are baseline point-estimate frequencies for overlapping 12-month changes. The intervals reflect the market weights only. Where the market reading is itself material (at least a quarter point, 274 months), the spending reading has the same sign in 94.5 percent of them and is within a quarter point in 58 percent. A practitioner would more often have sized a tightening differently than called an easing a tightening.

1998–2001. Averaged over the episode, the market-implied reading of the 12-month changes was slightly easing, at −0.06 points (interval −0.37 to 0.27), and the spending-equivalent reading was slightly tightening, at 0.16 (0.14 to 0.17). The difference of 0.22 (−0.12 to 0.53) is a point estimate; its interval includes zero, and so do the intervals for the separate contributions of the dollar (−0.05 to 0.36) and of equities (0.00 to 0.10). The signs differ in the averages and in 7 of the 48 months, and the monthly paths show what the average conceals (Figure 6). The difference was largest in the 12-month changes ending in the first half of 1998, at 0.76 points in January 1998, because those changes include the dollar's appreciation and the rise in valuations of 1997, and the model gives the dollar more weight and equities less. It shrank through 1999 and 2000 and turned negative in the second half of 2001 as valuations fell. The disagreement was therefore mostly about size and not direction: the signs differ in only 7 of 48 months, but the readings differ by at least a quarter point in 21. The dollar appreciated by 4.18 log points a year on average, which adds 0.16 to D; equity valuations rose by 2.70 log points a year, which adds 0.05. Unemployment rose by 0.35 points over the following year, against forecasts of 0.67 from the market-implied reading and 0.78 from the spending-equivalent reading.

The full indexes differ much more in levels. The FRB/US index read 1.79 points tighter than the market index on average (interval 0.33 to 2.97), and 1.32 of that difference comes from the

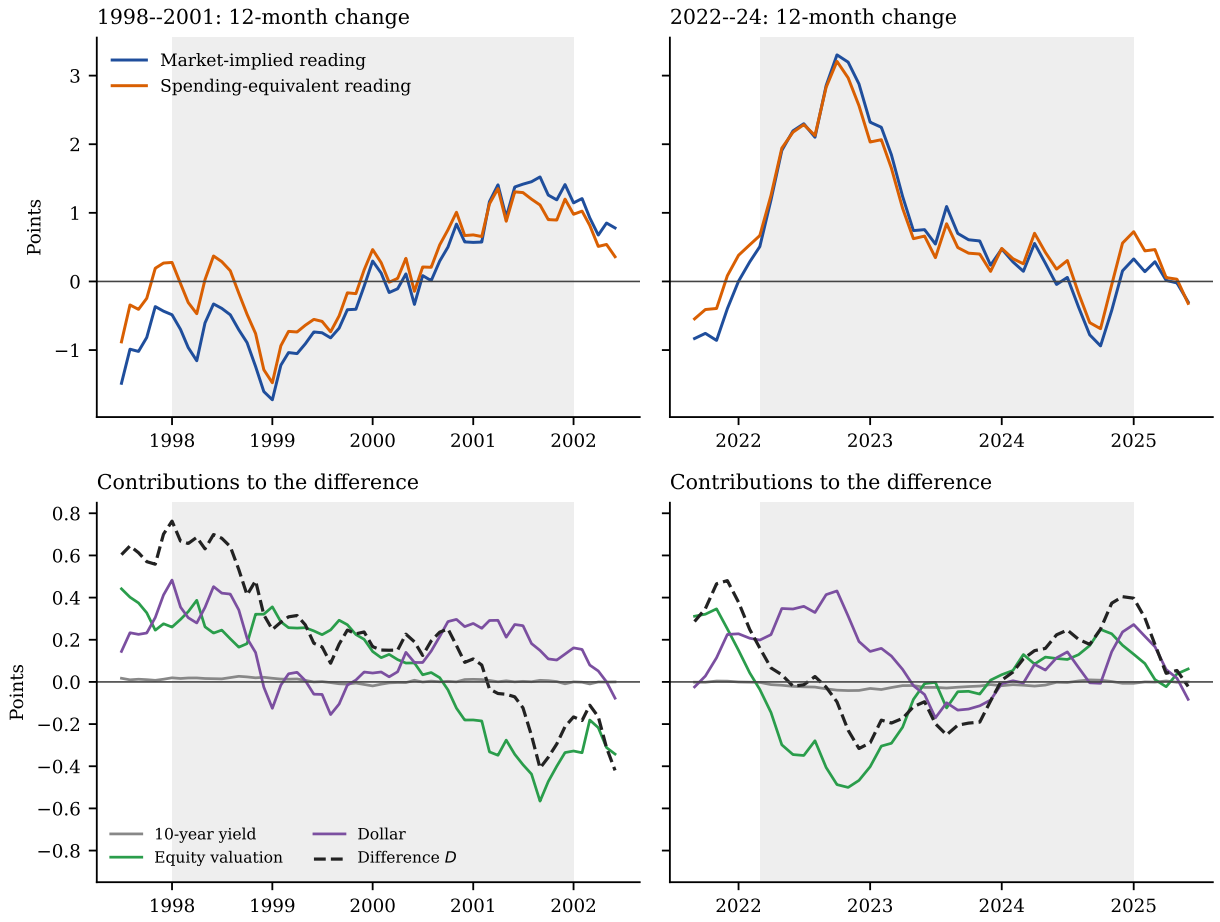


Figure 6: Month by Month: Readings and the Sources of Their Difference

Notes: Top: 12-month change of the market-implied and spending-equivalent readings on the same three assets relative to r^*. Bottom: each asset's contribution to D, the weight difference times its 12-month change, and D itself (dashed). The shaded area is the episode.

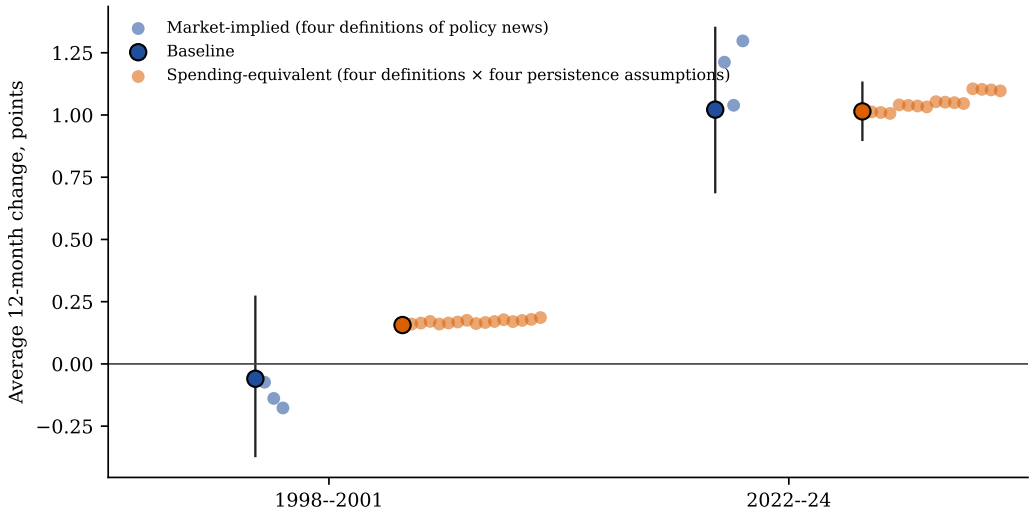


Figure 7: Two Readings of Two Episodes under Every Measurement Choice

Notes: Average 12-month change over the months of the episode on the same three assets relative to r^*. Market-implied: the four definitions of policy news of Table 3, in points of the two-year yield. Spending-equivalent: the same four definitions, which enter through the normalization, crossed with persistent paths and common half-lives of 8, 4 and 2 quarters, in funds-rate points. Large markers are the baseline. These are retrospective estimates. The 90 percent intervals of the baseline readings are in the text.

equity weights. Valuations were far above their average, and the market index, which gives each point of valuation more weight, read conditions as easier. The level and the change are different objects. The level difference is the weights applied to a persistent departure of valuations from their average; the difference in annual changes is the weights applied to what moved that year. Equities explain the first and the dollar explains most of the second.

2022–24. The two readings of the 12-month change averaged 1.02 points (interval 0.69 to 1.35) and 1.01 (0.90 to 1.13). Both movements tightened conditions. Equity valuations fell by 5.04 log points a year, and the dollar appreciated by 2.85. Because the model gives equities less weight than the event study, the fall in equities makes the spending-equivalent reading less tight than the market-implied one, and contributes −0.10 to D. Because the model gives the dollar more weight, the appreciation makes the spending-equivalent reading tighter, and contributes 0.11. The two contributions to the disagreement cancel, so that D averages −0.01 (−0.27 to 0.26), although each asset tightened conditions. The monthly paths show that the cancellation is not exact at every date: D ranged from about −0.3 to +0.5 within the episode, and it exceeded 0.25 in absolute value in 6 of 34 months. The baseline cancellation is specific to that definition of policy news: with the two-year yield as the signal, D is −0.17 and the market-implied reading is higher, and without the lower-bound years D is −0.19 (Section 3.2). A policymaker would have described a sizable tightening under every one of them. The full indexes differ in average level by 0.96 points (interval 0.00 to 1.76), with the equity weights (0.54) and the dollar (0.52) both contributing, and that difference does not affect the readings of the year's tightening. Unemployment rose by 0.27 points

over the following year, against forecasts of 0.65 from the market-implied reading and 0.66 from the spending-equivalent reading. Both weightings overpredicted the rise, so the weights do not explain the miss.

Across measurement choices. In 1998–2001 the market-implied readings run from −0.18 to −0.06 across the four definitions of policy news and the spending-equivalent readings from 0.16 to 0.19 across definitions and persistence assumptions, so that the ranges of the point estimates do not overlap and the signs differ throughout; the intervals of the baseline readings do overlap (Table 6 and the text above). In 2022–24 the market-implied readings run from 1.02 to 1.30 and the spending-equivalent readings from 1.01 to 1.11; both are about one point or more of tightening, and the ranges overlap. Persistence barely moves the normalized spending-equivalent reading, because the normalization absorbs the scale (the spread across persistence assumptions is 0.008 in 2022–24), although it moves the conversion of the typical policy response into funds-rate points. The definition of policy news moves the market-implied reading, and in 2022–24 the spread of the market-implied readings is larger than the spread of the spending-equivalent ones. On the market side the definition of policy news sets the range of the readings, and on the model side persistence adds little.

2004–06. The matched readings are close (the difference is −0.09), and the gap in levels is −0.11 (interval −0.76 to 0.41). The equity weights add 0.67 and house prices subtract 0.72, so the full FRB/US index contains a variable that the market index does not have, and equities do not explain the gap.

How representative are the episodes? I chose the episodes for their economic interest. They illustrate a mechanism and are not a random draw. By the monthly counts of Table 7, 1998–2001 is a period of unusually frequent large gaps (21 of 48 months against 36.6 percent of all months) and 2022–24 is quieter than typical (6 of 34 months).

These episodes give a concrete answer to when the choice of weights matters, in point estimates. The model gives less weight than the event study to equities and more to the dollar under the baseline definition, so the two readings part when valuations and the dollar rise together, as in 1998–99, and they offset when valuations fall as the dollar rises, as in 2022–24. The offset does not survive every definition of policy news, as the paragraph on measurement choices above shows.

6 Can Subsequent Activity Tell the Weightings Apart?

6.1 The Test

Consider two years in which the market index tightened by the same amount. If the dollar accounted for more of the tightening in one year, did activity subsequently weaken more? If the market-implied reading is a sufficient summary of the asset prices, the answer is no. If the spending-equivalent reading is, the answer is yes, by the amount that the model's weights imply. I regress the 12-month-ahead change in the unemployment gap on own lags and

$$y_{t+12} - y_t = \cdots + \beta M_t + \gamma D_t, \tag{4}$$

Table 8: Can Realized Activity Tell the Two Weightings Apart?

	n	Market index β [90%]	Composition γ [90%]	$\gamma - \beta$ [90%]
Unemployment gap, 12 months: 1976–2026	556	0.19 [0.05, 0.33]	−0.26 [−0.94, 0.43]	−0.45 [−1.01, 0.11]
Unemployment gap, 12 months: 1996–2026	318	0.01 [−0.21, 0.22]	−1.19 [−2.50, 0.13]	−1.19 [−2.31, −0.07]
Unemployment gap, 12 months: Excluding 2008–15	436	0.29 [0.15, 0.43]	0.32 [−0.13, 0.77]	0.03 [−0.30, 0.37]
Real GDP growth, four quarters: 1976–2026	186	−0.16 [−0.47, 0.16]	0.87 [−0.78, 2.51]	1.03 [−0.39, 2.44]
Real GDP growth, four quarters: 1996–2026	106	0.31 [−0.13, 0.75]	1.81 [−0.76, 4.38]	1.50 [−0.70, 3.71]
Real GDP growth, four quarters: Excluding 2008–15	146	−0.33 [−0.61, −0.04]	0.26 [−1.21, 1.72]	0.58 [−0.69, 1.86]

Notes: Local projections of the 12-month-ahead change in the unemployment gap (four-quarter-ahead GDP growth) on own lags and $y = \cdots + \beta M + \gamma D$, where M is the 12-month change of the market index relative to r^* and D the 12-month change of the normalized FRB/US reading less the market reading, computed on the same changes (HAC standard errors, 18 lags, joint covariance of the coefficients). If the market reading is right, $\gamma = 0$. If the normalized FRB/US reading $M + D$ is right, $\gamma = \beta$. A failure to reject is not evidence for either.

where M is the 12-month change in the market index (relative to r^*) and D the 12-month change of the normalized FRB/US-minus-market difference on the same changes. If the market-implied reading is sufficient, $\gamma = 0$. If the spending-equivalent reading $M + D$ is sufficient, $\gamma = \beta$. I read the two restrictions as tests of incremental predictive sufficiency in the span of the two portfolios, conditional on the controls. Three assets span a larger space, and the unrestricted three-asset regression of Section 6.3 is a different test. Failure to separate the portfolios for aggregate unemployment or GDP is consistent with low power and with forward-looking asset prices, and it cannot validate either structural weight. That wording matters for unemployment, since the FRB/US weights are constructed from effects on GDP; four-quarter GDP growth is the model-facing check. This is the only primary test, and Table 8 reports both restrictions with the joint covariance of the estimates.

6.2 The Answer

On 1976–2026 the coefficient on the index change is 0.19 and on composition −0.26 (−0.94 to 0.43); the difference $\gamma - \beta$ is −0.45 (−1.01 to 0.11). Neither restriction is rejected. Moving D from its 5th to its 95th percentile at the same index change shifts the year-ahead outlook by −0.29 percentage point, with an interval from −1.06 to 0.48, so the sample cannot establish an outlook difference of either sign. On 1996–2026 the coefficient on composition is −1.19 (−2.50 to 0.13) and the model's restriction is rejected ($\gamma - \beta = -1.19$, −2.31 to −0.07); excluding 2008–15 the coefficient on composition is 0.32 and $\gamma - \beta$ is 0.03, close to the model's restriction. The sign changes across samples. For GDP growth neither restriction is rejected, and adding D does not significantly improve out-of-sample forecasts (Clark–West −1.8).

6.3 Equities Alone

Equity valuation alone raises the adjusted R^2 of the change in unemployment by 0.115, the market index by 0.136, the FRB/US weights by 0.114, the 10-year yield by 0.099, the dollar by 0.004 and the three variables entered freely by 0.147. Both indexes draw heavily on equity valuation, which helps explain why unemployment data have difficulty distinguishing between them: the movements that

Table 9: Composition and the Expenditure Components of Growth

	n	Coefficient [90% HAC]	Bootstrap 90%	Effect, 5th to 95th percentile
Net exports, 1976–2026: index change β	199	−0.11 [−0.19, −0.02]	[−0.19, 0.00]	−0.11
Net exports, 1976–2026: composition γ	199	−0.74 [−1.06, −0.41]	[−1.04, −0.37]	−0.79
Consumption and private investment, 1976–2026: index change β	199	−0.19 [−0.54, 0.15]	[−0.52, 0.22]	−0.21
Consumption and private investment, 1976–2026: composition γ	199	1.71 [−0.29, 3.70]	[−0.35, 3.92]	1.83
Consumption and investment less net exports, 1976–2026: γ	199	2.44	[0.13, 4.88]	2.63
Net exports, 1996–2026: index change β	119	0.03 [−0.13, 0.19]	[−0.13, 0.24]	0.03
Net exports, 1996–2026: composition γ	119	−0.54 [−1.17, 0.09]	[−1.28, 0.10]	−0.48
Consumption and private investment, 1996–2026: index change β	119	−0.31 [−1.10, 0.48]	[−1.13, 0.72]	−0.27
Consumption and private investment, 1996–2026: composition γ	119	2.95 [−0.65, 6.55]	[−0.98, 6.11]	2.59
Consumption and investment less net exports, 1996–2026: γ	119	3.49	[−0.99, 7.17]	3.07

Notes: Local projections of the mean contribution to real GDP growth (percentage points, annualized) of net exports and of consumption plus private investment over the next four quarters on own lags, the 12-month change of the market index and the 12-month change of D (Section 6.1); quarterly origins, HAC standard errors with 6 lags, 8-quarter block bootstrap with common draws for the two outcomes. Predictions stated from the channel logic before the components were estimated: the coefficient on D is negative for net exports (P1) and larger for domestic demand than for net exports (P2). Effect: coefficient times the 5th-to-95th percentile range of D at a given index change.

distinguish the weightings are the ones in which the dollar and equity valuation move in different proportions, and there are few of them. Entered freely, the coefficients are not significantly different from proportional to the market weights ($p = 0.53$) or to the FRB/US weights ($p = 0.22$); the power is low.

6.4 Trade-Sensitive and Domestic Expenditure

Section 3.3 traced the model's weights to channels: consumption and investment for equities, net exports for the dollar. If the weights describe those channels, composition might predict the components separately even where it does not predict the aggregates. I wrote two predictions from the channel logic before estimating the components: the coefficient on D is negative for the contribution of net exports to growth over the next four quarters (P1), and it is larger for consumption and private investment than for net exports (P2). The aggregate nulls of the preceding subsections were already known, so the exercise is a check on a prediction and not a clean pre-registered test. Table 9 reports both.

In 1976–2026 the coefficient on D for net exports is −0.74 (−1.06 to −0.41; bootstrap interval −1.04 to −0.37), an effect of −0.79 percentage points of growth between the 5th and 95th percentiles of D. For consumption and private investment it is 1.71 (−0.29 to 3.70), and the difference has a bootstrap interval of 0.13 to 4.88. Both predictions hold in the baseline main sample. They do not survive further scrutiny as evidence for the dollar channel. In this sample D is dominated by equities: it correlates 0.86 with the 12-month change in equity valuation and 0.31 with the change in the dollar. Splitting D into the parts due to each asset, the equity part has a coefficient of −0.97 (−1.67 to −0.27) and the dollar part −0.52 (−1.20 to 0.16). Rising equities strengthen domestic demand and imports, which lowers the net-export contribution mechanically. With the three asset changes entered freely, the dollar's own coefficient is negative, as the dollar channel implies (−0.030, −0.050 to −0.011), and so is the equity coefficient (−0.010). On 1996–2026 the

Table 10: Composition and Outcomes Under Each Definition of Policy News

	Unemployment, γ	$\gamma - \beta$	GDP growth, γ	$\gamma - \beta$	Net exports, γ
Futures factor	−0.26 [−0.94, 0.43]	−0.45 [−1.01, 0.11]	0.87 [−0.78, 2.51]	1.03 [−0.39, 2.44]	−0.74 [−1.06, −0.41]
Two-year yield	−0.10 [−0.50, 0.31]	−0.33 [−0.75, 0.09]	0.53 [−0.51, 1.57]	0.77 [−0.25, 1.78]	0.02 [−0.37, 0.41]
Factor, through 2019	−0.12 [−0.64, 0.39]	−0.31 [−0.71, 0.09]	0.54 [−0.67, 1.76]	0.71 [−0.29, 1.70]	−0.54 [−0.78, −0.30]
Factor, no lower bound	−0.19 [−0.57, 0.20]	−0.38 [−0.72, −0.03]	0.68 [−0.35, 1.72]	0.85 [−0.07, 1.77]	−0.24 [−0.55, 0.07]

Notes: Coefficient on the composition term D (and its difference from the coefficient on the market index change) in the regressions of Sections 6.1 and 6.4, 1976–2026, with D and the market index change computed with the market weights of each definition of policy news and the model weights normalized by that definition's $b_4'\psi$; HAC 90 percent intervals.

signs are the same and the evidence is weaker: the net-export coefficient is −0.54 (−1.17 to 0.09). The result also depends on the definition of policy news (Table 10): the coefficient on D for net exports is −0.74 for the baseline, −0.54 through 2019, −0.24 without the lower-bound years and 0.02 with the two-year yield, where it is indistinguishable from zero. Composition predicts net exports under the futures-factor definitions, mostly through equities. It does not provide clean evidence for the dollar channel of the model.

Table 10 also repeats the unemployment and GDP regressions under each definition. The coefficient on composition is not significantly different from zero in any of them, and the model's restriction $\gamma = \beta$ is rejected at the 10 percent level for unemployment only without the lower-bound years.

6.5 What Forecasters Expected and What Happened

Forecasters' errors combine the outcome with what they had already anticipated. Table 11 regresses the Survey of Professional Forecasters' unemployment forecast, the realized change and the error (realized less forecast), over the same 128 surveys (1990Q3–2025Q3), on the index change and D. The coefficients on the forecast and the error add to the coefficient on the realization. When FRB/US reads the same asset movements as tighter than the market index does, the forecast does not respond (−0.18, interval −0.60 to 0.24), unemployment tends to come in lower (−1.47, −2.83 to −0.11), and so the error is negative (−1.29, −2.37 to −0.21). Composition predicts what subsequently happened, not what forecasters expected, and with a sign opposite to the one the spending-equivalent reading would imply if forecasters had not priced composition. The result depends on the sample, and I test which episodes carry it. Excluding 2001 changes little (the error coefficient is −1.45, interval −2.58 to −0.32). Excluding 2008–09 reduces it to −0.37 (−0.69 to −0.04) and the coefficient on the realization to −0.28 (−0.84 to 0.27). Excluding 2008–15 the forecast coefficient is 0.29 and the realization −0.18, and the error is −0.46, with a robust interval of −0.88 to −0.05, which excludes zero, and a block-bootstrap interval of −0.76 to 0.20, which includes it. Without both 2001 and 2008–09 the error is −0.44 (−0.78 to −0.10). The sign of the error coefficient is stable and its size is not: most of the full-sample effect on the realization comes from 2008–09. The outcomes overlap and the coefficients are not adjusted for their number.

Table 11: What the Composition Predicts: Forecasts, Outcomes and Errors

	n	Coefficient on D [90%]	Effect, 5th to 95th percentile
Unemployment, 1990–2025: Survey forecast	128	−0.18 [−0.60, 0.24]	−0.16
Unemployment, 1990–2025: Realized	128	−1.47 [−2.83, −0.11]	−1.33
Unemployment, 1990–2025: Forecast error	128	−1.29 [−2.37, −0.21]	−1.17
Unemployment, excluding 2008–15: Survey forecast	88	0.29 [−0.06, 0.64]	0.25
Unemployment, excluding 2008–15: Realized	88	−0.18 [−0.83, 0.48]	−0.15
Unemployment, excluding 2008–15: Forecast error	88	−0.46 [−0.88, −0.05]	−0.40
Real GDP growth, 1990–2025: Survey forecast	128	−0.35 [−0.93, 0.22]	−0.32
Real GDP growth, 1990–2025: Realized	128	2.15 [−0.05, 4.35]	1.95
Real GDP growth, 1990–2025: Forecast error	128	2.51 [0.50, 4.51]	2.27
Unemployment, excluding 2001: Realized	116	−1.69 [−3.10, −0.28]	−1.58
Unemployment, excluding 2001: Forecast error	116	−1.45 [−2.58, −0.32]	−1.35
Unemployment, excluding 2008–09: Realized	112	−0.28 [−0.84, 0.27]	−0.24
Unemployment, excluding 2008–09: Forecast error	112	−0.37 [−0.69, −0.04]	−0.30
Unemployment, excluding 2001 and 2008–09: Realized	100	−0.40 [−0.99, 0.20]	−0.34
Unemployment, excluding 2001 and 2008–09: Forecast error	100	−0.44 [−0.78, −0.10]	−0.37

Notes: Regressions of the Survey of Professional Forecasters' median forecast of the four-quarter-ahead change in the unemployment rate (real GDP growth), the realized change, and the forecast error (realized less forecast) on the 12-month changes of the market index and of D, financial data up to the month before each survey's deadline, 1990Q3–2025Q3; the coefficients on the forecast and the error add to the coefficient on the realization. HAC standard errors; effect in percentage points at the same index change. Incremental information only: the test says nothing about causation.

What can a policymaker take from this? The two readings can be used interchangeably when markets move as they do around policy news, and the data cannot say which to prefer when they do not. A reading from either index should be accompanied by the question of what moved: the dollar, valuations, or yields. The answer indicates whether the two readings are likely to part.

7 Conclusion

The paper asks how many points of monetary policy a movement in financial conditions is worth. The answer depends on the question. An event study says how large a conventional policy surprise the movement resembles. A spending model says how large a policy-rate change would have a comparable effect on GDP. For the usual response to a policy surprise the two are close once they are put on one scale, and that scale is a measurement choice: the model prices the response at 0.954 funds-rate points with persistent paths and 0.82 with a half-life of four quarters. For other combinations of asset prices the readings differ. In point estimates the model gives the dollar more weight, relative to equities, than the event study does under every definition of policy news that I try. The model's weights come from two channels, consumption and investment for equities and net exports for the dollar.

The second result is that the measure of policy news matters about as much as the choice between market and spending weights. In 12-month changes the average difference between the model and the baseline market reading is 0.33 of the average market move, and the average difference between the baseline market reading and the market reading with the two-year yield is 0.37. The

sampling noise in the market weights is of similar size, so the data cannot tell the calibrations apart much more finely than that.

The third result is how often this matters. Across 369 months from 1996 to 2026 the baseline readings agree on the direction of the annual change in 87.5 percent of months, differ by at least a quarter point in 36.6 percent, and oppose each other materially in only 8. In 1998–2001 the readings differed mostly in size, with a gap of about 0.7 points in early 1998. In 2022–24 every calibration shows a large tightening of about one point or more. The baseline offset of the equity and dollar contributions is specific to that definition of policy news.

Aggregate activity does not separate the two readings, and the tests have little power. After 1996 the model's restriction is rejected for unemployment in the baseline sample, and the forecast-error coefficients of Section 6.5 are significant for the full sample and depend on 2008–09. Composition predicts the net-export contribution to growth under the futures-factor definitions, mostly through equities, which is not clean evidence for the dollar channel. A movement that departs from the policy-news mix can come from the announcement itself (Boehm and Kroner, 2025).

The limits are those of the design. The FRB/US weights come from one model and one experiment and are held fixed in the intervals. The market weights are imprecise, the horizons are compared on overlapping data, the daily comparison uses nominal assets, and there are five recessions in the sample. The paper does not say which definition of policy news is right. A reader who wants to use either index can report both readings and can treat their difference as a measure of how much the answer depends on the calibration.

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Appendix A The Weights: a Derivation

Let the movements in the three asset prices be $x = \psi s + u$, where s is policy news, ψ the responses and u everything else, with covariance Σ. Let g_h be the effects of a persistent unit shift in each asset price on spending at horizon h. The FRB/US weights are proportional to g_h. The market weights solve the problem of recovering s from x: the generalized least squares estimator is $\hat{s} = w_M' x$ with $w_M = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$, which satisfies $w_M'\psi = 1$ and, by the Gauss–Markov theorem, minimizes the variance of the estimate among linear unbiased estimators of s when u has mean zero and covariance Σ and is uncorrelated with s. For any x the difference between the normalized readings is $D_h(x) = (w_{B,h} - w_M)'x = (w_{B,h} - w_M)'u$, because the policy part cancels. The readings coincide for every movement if and only if $w_{B,h} = w_M$, that is, if $g_h \propto \Sigma^{-1}\psi$: the criterion that recovers policy news precisely and the criterion that measures spending effects select the same weights only in that case. The variance of the difference is $(w_{B,h} - w_M)'\Sigma(w_{B,h} - w_M)$.

Observed movements need not be of the form $\psi s + u$ with a single policy factor, and the model's effects need not be those of a movement of the same shape. If all observed movements lay along the single direction ψ, every weight vector normalized to price ψ at one would give the same reading on them. The data would then identify the reading along that direction and leave the weights themselves undetermined. In the announcement windows u is not zero, and it is the variation in u that identifies Σ and hence the weights. What distinguishes the two weightings in practice is the exposure of the difference in weights to the movements that occur, which Table A.1 measures as a ratio to the market move.

Appendix B Construction of the Series

The three asset prices enter in three forms, each answering a problem of the one before. The headline results use the third.

Nominal. Rates as published. A nominal index treats a 10-year yield of 8 percent alike in 1981 and in 2000. That is reasonable only if inflation expectations and the neutral real rate are stable,

Table A.1: How Large is the Difference Between the Two Readings?

Movements	Mean $\lvert D\rvert$	Ratio to market move [90%]	Share above 0.25 (%)	Mean run (months)
Announcement windows, held out	0.010	0.19 [0.09, 0.37]	0	–
Monthly changes, 1996–2026	0.043	0.24 [0.14, 0.54]	0	1.0
Monthly changes, 1973–2026	0.063	0.27 [0.15, 0.51]	2	1.1
12-month changes, 1996–2026	0.26	0.33 [0.17, 0.63]	37	7.5
12-month changes, 1973–2026	0.33	0.32 [0.16, 0.54]	45	7.3

Notes: D is the FRB/US reading (four quarters, normalized so that a conventional policy surprise reads one) less the market reading, applied to the changes of the equity valuation, the 10-year yield and the dollar at the r^* step. Announcement windows: held-out events (weights re-estimated without each event). Ratio: mean $\lvert D\rvert$ over mean absolute market move; interval from bootstrap samples of the announcements and, for the monthly history, of 12- or 24-month blocks of months. A run is consecutive months with $\lvert D\rvert$ above 0.25.

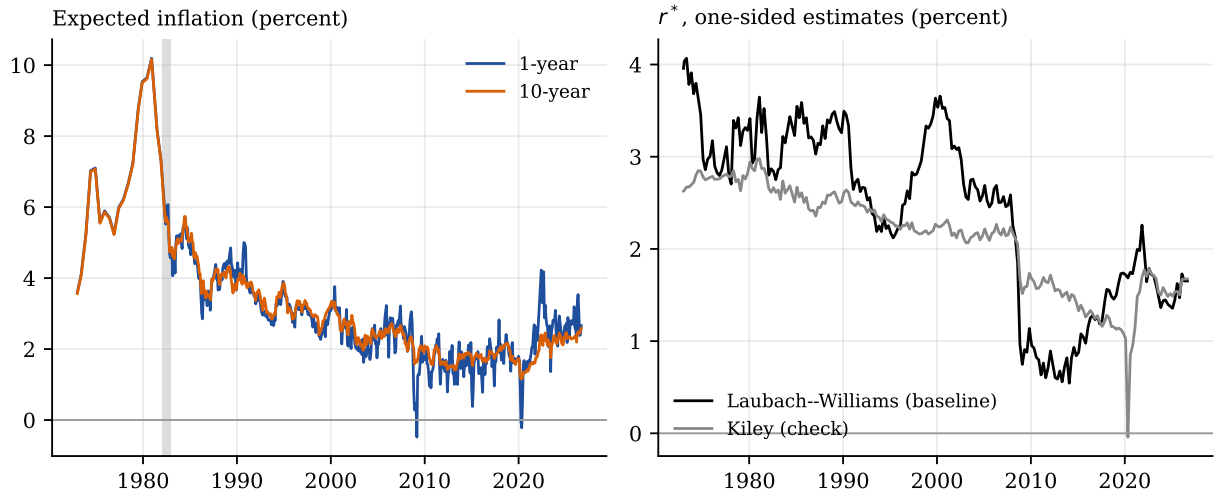


Figure B.1: Expected Inflation and r^*

Notes: Left: expected inflation, Livingston Survey before 1982 and Cleveland Fed model estimates after, shifted to the Cleveland level over 1982–84 and blended over 1982; shaded: the blending year. Right: one-sided, full-sample estimates of r^* from the Laubach–Williams model (baseline) and Kiley's model (check). Percent.

and neither was. Expected inflation over the next year averaged 8.43 percent in 1981 and is 2.66 percent today, and the Laubach–Williams estimate of r^* averaged 2.76 percent over 1990–2007 and 0.86 percent over 2009–2016 (Figure B.1).

Real. Each rate less expected inflation at a matched horizon: ten-year expected inflation for the 10-year yield. The dollar is the real broad index.

Relative to r^*. The real 10-year yield less r^*. Equity valuation enters through the earnings yield $E/P = 100/\text{CAPE}$, which is also an implicit real rate, as the gap $E/P - r^*$, converted to CAPE log points at the sample-average earnings yield so that its weight on CAPE is unchanged. The baseline uses the one-sided, full-sample Laubach–Williams estimate. Each variable is measured relative to its 1973–2026 average, so r^* matters through its movements over time.

The transformation matters for levels and not for the difference between the two readings. The ratio of the mean absolute difference to the mean absolute market move in 12-month changes, 1996–2026, is 0.33 (interval 0.18 to 0.68) with nominal assets and 0.33 relative to r^*.

Data. All series are monthly averages from January 1973 to September 2026, from the public sources in Table B.1. The BBB spread is Moody's Baa yield less the 10-year yield. Months in which the CPI and unemployment releases are missing (October 2025) are interpolated. The unemployment gap is the unemployment rate less the Congressional Budget Office's natural rate, placed in the middle month of each quarter and interpolated. Expected inflation is the Cleveland Fed's estimate at one and ten years from 1982 and the Livingston Survey's 12-month CPI expectation before, shifted to the Cleveland level by the mean difference over 1982–84 and blended over 1982. CAPE is Shiller's series spliced over 1990–92 in logs to the monthly average of a daily series. The real broad

dollar is spliced in 2006 in logs, as are the nominal dollar and the S&P 500.

Market index. Policy news is the first principal component of MP1, MP2, ED2, ED3 and ED4 in the monetary-event windows, scaled by its regression on the two-year yield change so that one unit moves the two-year yield by one percentage point. Unscheduled events with no change in MP1 and March–April 2020 are dropped. Each of the 10-year yield change, the S&P 500 return and the dollar (DXY) return in the window is regressed on the factor. The residual covariance Σ of the three regressions gives $w = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$. The S&P 500 and DXY weights apply to 100 times the log of CAPE and of the real broad dollar, as percent changes in the window and in the index levels agree to first order. The weights bootstrap resamples announcements with replacement and recomputes the policy-news factor and the weights each time; draws use the seed in `lib/config.py`.

FRB/US index. The weights are the cumulative GDP responses at four quarters to a persistent unit shift in each variable, simulated in FRB/US with VAR-based expectations, divided by the response to a persistent one percentage point rise in the funds rate. The weight on the policy rate is its direct effect with expectations of future rates held fixed. House prices enter as 100 times the log of the price-to-rent ratio relative to its trailing 15-year average. The expenditure decomposition of Figure 3 uses the model's own identities for consumption, residential and business fixed investment, inventories, government, exports and imports, weighted by their shares in GDP, with a residual below 10^{-7} percentage points; it is run with the Board's FRB/US package and the data vintage of October 2026, downloaded by step 00 into a gitignored directory, with a Python 3.11 environment built on first use. The policy-rate experiment shocks the inertial-Taylor-rule equation in the first quarter and is scaled (to 2.14 units of the add factor) so that the average simulated funds rate over eight quarters rises by one point; GDP at four quarters is linear in the size of the shock to within 1.6 percent. For the decay experiments of Section 3.5, each variable's shock decays geometrically with the half-life stated, and the response is computed in the model by linearity from the temporary-shock responses.

Policy-news definitions. The two-year definition replaces the futures factor by the change in the two-year Treasury yield in the same window, with the same regressions and the same bootstrap of announcements; the draws for different definitions are paired by keeping, for each resample, the announcements that fall in each subsample, so the subsample size varies by draw. The announcement bootstrap uses 5,000 draws for every weights-based interval; the block bootstraps of the VAR and the outcome regressions use 500. The FRB/US weights are held fixed.

Forecasting and survey data. The regressions use local projections of the 12-month-ahead change in the unemployment gap (four-quarter-ahead GDP growth) with HAC standard errors of 18 lags. The Survey of Professional Forecasters data are the median forecasts, 1990Q3–2025Q3, matched to the financial data up to the month before each survey's deadline.

Table B.1: Data Sources

Series	Frequency	Source
Federal funds rate; 10-year yield; Baa yield; 30-year mortgage rate	Monthly	FRED FEDFUNDS, GS10, BAA, MORTGAGE30US
Real broad dollar; nominal broad dollar	Monthly	FRED RTWEXBGS, TWEXBPA; DTWEXBGS, TWEXBMTH (spliced in 2006)
CPI; CPI rent; Case–Shiller home prices; S&P 500	Monthly	FRED CPIAUCSL, CUUR0000SEHA, CSUSHPISA, SP500; Shiller's data before the FRED start
Unemployment rate; natural rate of unemployment; real GDP	Monthly; quarterly	FRED UNRATE; CBO, FRED NROU; GDPC1
Expected inflation, 1 and 10 years	Monthly	Cleveland Fed, FRED EXPINF1YR, EXPINF10YR; before 1982, Livingston Survey (Philadelphia Fed)
r^*, Laubach–Williams	Quarterly	New York Fed, one-sided estimates
r^*, Kiley	Quarterly	One-sided estimates of the author
CAPE	Monthly	Shiller, spliced to the monthly average of a daily CAPE constructed by the repository's daily workflow
Survey of Professional Forecasters	Quarterly	Federal Reserve Bank of Philadelphia
FRB/US simulations	Quarterly	Run for this paper's companion work with the public FRB/US package
Futures, yields, S&P 500 and DXY in the window	Event	U.S. Monetary Policy Event-Study Database, "monetary events" (Acosta et al., 2025)

Appendix C Inference and the Decomposition of the Levels

C.1 How Precisely Are the Weights Estimated?

The weights come from regressions on 253 announcements. The weight on the 10-year yield has a 90 percent interval from 0.15 to 1.11 around 0.61. The interval for equity valuation runs from -0.07 to -0.03, and for the dollar from 0.02 to 0.11. The FRB/US weights on all three lie inside these intervals, the equity weight near the edge: it is lower in absolute value than 5 percent of the bootstrap draws, about 14 percent in a two-sided test. The data do not reject that the model weights equal the market weights. This is a statement about power, not a finding that they are equal, and it treats the FRB/US weights as fixed. The intervals in the text come from the bootstrap of the announcements; the contrast in composition in Section 3.5 is the ratio of ratios, whose interval includes one at two quarters.

C.2 The Two Tightenings of Figure 1

Table C.1 gives the intervals behind Figure 1.

Table C.1: Two Financial Movements that the Market Index Rates Alike

	Market index	90% interval	FRB/US index
Equity valuation falls 5 log points	0.256	[0.152, 0.342]	0.153
Dollar appreciates 3.9 log points	0.255	[0.081, 0.438]	0.384
Dollar effect over equity effect	1.00	[0.33, 2.25]	2.51
Typical joint response to a unit of policy news	1.00		0.954 [0.86, 1.08]

Notes: Weights applied to the stated movements, with no change in the 10-year yield. Market index: two-year-yield points per movement, weights from announcement-window responses (Table 1); the interval takes the 5th and 95th percentiles over bootstrap samples of the announcements. FRB/US index: funds-rate points; the effect on GDP after four quarters of a persistent shift in the variable, divided by the effect of a persistent one percentage point rise in the funds rate. Typical joint response: the index applied to the responses of the three variables to a unit of policy news (one by construction for the market index).

Table C.2: From the Market Index to the FRB/US Index: Shares of the Difference

Stage	Level, share of variance (%)	12-month change (%)
Equity weights	72 [0, 89]	53 [-4, 87]
Dollar weights	12 [-1, 38]	15 [-1, 34]
10-year weight	1 [-4, 51]	1 [-7, 55]
Normalization of the weights	5 [-4, 19]	4 [-2, 16]
Policy rate	-0 [-1, 0]	0 [-1, 2]
BBB spread	1 [-1, 8]	3 [-3, 15]
Mortgage spread	2 [-3, 15]	15 [-0, 43]
House prices	7 [-2, 34]	8 [0, 24]

Notes: Shares of the variance of the FRB/US index (with house prices) less the market index, relative to r^*, 1973–2026, attributed to each stage of a bridge between them: the weights on the three common assets, the normalization of the weights, then the contributions that only the FRB/US index has (policy rate, spreads, house prices); both indexes use the same inputs and centering. Intervals (5th to 95th percentile) from bootstrap samples of the announcements and of blocks of months.

C.3 From the Market Index to the FRB/US Index

Table C.2 accounts for the difference in average levels between the full FRB/US index and the market index, in stages on the same inputs and centering: the variables the market index does not have (the policy rate, spreads and house prices), the equity weights, the dollar weights and the 10-year weights.

Appendix D Sensitivity

D.1 Weights That Move

The weights define the unit, so I hold them fixed and show how they would drift. Re-estimating on an expanding window of announcements from 2006 keeps the 10-year and dollar weights inside the 90 percent band of the full-sample estimate at every date after 2010; the equity weight is outside at 32 dates between January 2011 and January 2015, by at most -0.073 against a band edge of -0.068

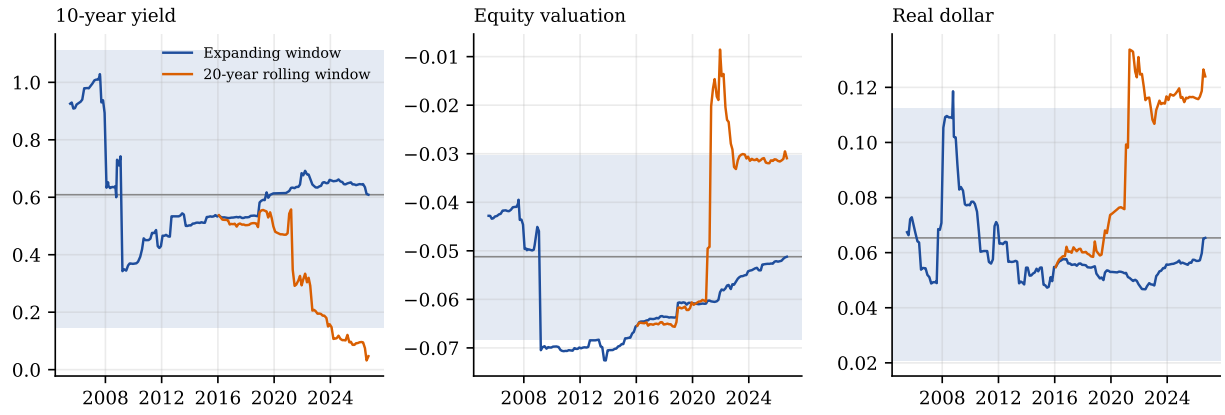


Figure D.1: The Weights on an Expanding and a Rolling Window

Notes: Weights of the market index re-estimated on announcements to each date (expanding window, from the 80th announcement) and on the latest 20 years (rolling). Shaded: 90 percent bootstrap interval of the full-sample weights; the horizontal line is the full-sample weight.

Table D.1: Weights Under Other Policy-News Series and Event Samples

	Events	w 10-year	w equities	w dollar	ψ 10-year	ψ equities	ψ dollar
Baseline: Nakamura–Steinsson factor	253	0.609	−0.051	0.065	0.53	−7.78	4.22
Policy news is the 2-year yield change	253	0.925	−0.027	0.070	0.58	−6.09	4.22
Announcements through 2019	201	0.613	−0.061	0.053	0.56	−7.50	3.83
No announcements December 2008–November 2015	197	0.986	−0.039	0.058	0.49	−7.82	3.69

Notes: Weights w and responses ψ as in Table 1, re-estimated on each sample or with the window change of the two-year yield as the policy-news measure. In each case the weights applied to the responses sum to 1.

(Figure D.1). A 20-year rolling window fails after 2021, because it drops the announcements of 1996–2005. The 10-year weight is imprecise throughout. Apart from the equity weight in 2011–15, the weights stay within their sampling error.

D.2 Other Policy-News Series and Event Samples

Table D.1 re-estimates the weights with the two-year yield as the policy news, with announcements through 2019, and without the lower-bound years. The equity weight is −0.027, −0.061 and −0.039 against −0.051 in the baseline, and the conclusions about composition rest on the model's ratio of dollar to equity effects, not on any one set of market weights.

D.3 The Policy-Rate Experiment in FRB/US

A policy shock in the model gives a comparison with the announcement response. I shock the equation of the inertial Taylor rule in the first quarter only, with the rule active afterwards and expectations from the model's VAR, and I scale the shock so that the average simulated funds rate over the next eight quarters rises by one point. FRB/US has no two-year yield, so that average is

my proxy for the unit of policy news. Per unit of the proxy, the 10-year yield rises 1.33 times, equity wealth falls 4.01 times and the dollar rises 0.55 times as much as in the announcement response. GDP four quarters later falls by -0.82 percent, a funds-rate equivalent of 1.77, against 0.95 for the usual announcement response. Priced along their paths, the 10-year yield, equities and the dollar account for 20, 43 and 13 percent of the GDP effect of the model's policy shock, against 32, 25 and 44 percent for the announcement response, and 23 percent is left in the policy rate, spreads and house prices. The model's own policy transmission is heavier in equities and lighter in the dollar than the announcement response, and the difference in the funds-rate equivalent, 1.77 against 0.95, is the residual between the two experiments. The comparison has limits. The funds-rate average is a proxy, and a different proxy changes the unit. The equity fall on impact is large because stock wealth in FRB/US follows expected funds rates through the required return. The exercise is the model's own policy shock, not a reproduction of an announcement. The reading of about one in Section 3.4 is therefore not the statement that announcements move asset prices as a policy shock does in the model; it is a statement about the combined effect on GDP of different mixes.

D.4 Limits of the Design

The weights come from announcements since 1996 and are applied from 1973; they assume that the responses reflect policy rather than information effects (Nakamura and Steinsson, 2018), and if policy came to target financial conditions the responses could change, the Lucas-critique concern that Hatzius et al. (2010) raise for all indexes. The FRB/US weights come from one model, at one horizon and in one experiment, and are held fixed in the intervals. The daily comparison of Section 4 uses nominal assets, with the nominal broad dollar for the real dollar, no adjustment for r^* and weights fitted on the same announcements; the intraday comparison holds each event out. The sample has five recessions, two of them out of sample, and the 12-month outcomes overlap, so the counts of tests are descriptive and not adjusted for their number. The neutral rate is not neutral to outcomes, because the Laubach–Williams model uses output, inflation and unemployment.